

Report: Optimising NYC Taxi Operations

New York City Yellow Taxi Data, 2023 Analysis

1. Data Preparation

1.1 Loading the Dataset

Sample the data and combine the files

The dataset consists of 12 monthly parquet files covering all of 2023 (approx. 37.9M rows total). Loading all files at once isn't feasible memory-wise, so a stratified 5% sample was taken from each hour of each date within every monthly file.

Sampling by both date AND hour is important. Sampling only by hour across the whole month could miss quiet mid-week days and over-represent busy Fridays. Stratifying by date×hour ensures the sample reflects the actual daily and intra-day demand patterns throughout the year.

The final combined dataset has 1,896,400 rows covering all 12 months of 2023.

2. Data Cleaning

2.1 Fixing Columns

Fix the index

After concatenating the 12 monthly files, the row index had duplicate values (each monthly file starts from 0). This was fixed with `reset_index(drop=True)` to get a clean sequential index.

Helper columns created during sampling (`pickup_date`, `pickup_hour`) were dropped as they're no longer needed. The `store_and_fwd_flag` column was also dropped, as it only records whether a trip record was held in vehicle memory due to connectivity issues, which isn't relevant to any analysis.

Combine the two `airport_fee` columns

When the 12 monthly parquet files were concatenated, two airport fee columns appeared in the dataframe: `airport_fee` (January's file uses this lowercase name) and `Airport_fee` (all other 11 months). This is a naming inconsistency in the source data.

The two columns were merged into a single `airport_fee` column. The merge was done by grouping columns by their lowercase names, keeping the variant with the most non-null values, and renaming to the lowercase standard.

2.2 Handling Missing Values

Find the proportion of missing values in each column

Missing values were found in the following columns:

```
passenger_count: 3.43%
ratecodeid: 3.43%
congestion_surcharge: 3.43%
airport_fee: ~11.0%
```

All other columns had no missing values. The ~3.43% pattern across three columns suggests these correspond to the same set of trips where vendor data was incomplete.

Handling missing values in `passenger_count`

NaN values in `passenger_count` were imputed with the median value (1.0). The median is preferred over the mean here since `passenger_count` is discrete and the distribution is heavily right-skewed (most trips have 1-2 passengers).

Zero values were also found and replaced, since a completed trip cannot physically have 0 passengers, so these are data entry errors. After imputation, `passenger_count` values of 7 and above were removed since NYC yellow taxis are not licensed for groups larger than 6.

Handle missing values in `RatecodeID`

Missing `RatecodeID` values were filled with the mode (1 = Standard rate), which accounts for the overwhelming majority of trips (1,727,956 out of ~1.9M). `RatecodeID` = 99 was also found in ~10,724 rows. This value is not defined in the data dictionary and was treated the same way.

Impute NaN in `congestion_surcharge`

NaN values in `congestion_surcharge` were filled with 0. A missing value here means the surcharge wasn't applied, most likely because the trip didn't pass through the designated NYC congestion zone. The `airport_fee` column was handled the same way for the same reason.

2.3 Handling Outliers and Standardising Values

Check outliers in payment type, trip distance and tip amount

Looking at the describe() output, several obvious outlier conditions stood out:

passenger_count > 6: removed. NYC taxis are legally limited to 6 passengers.

trip_distance < 0.1 miles AND fare_amount > \$300: removed. No trip under 0.1 miles can legitimately cost \$300+.

trip_distance = 0 AND fare_amount = 0 but different pickup/dropoff zones: removed. Different zones means there must be some distance and fare.

trip_distance > 250 miles: removed. NYC is about 13 miles end-to-end, so 250+ mile trips are not plausible.

payment_type = 0: not defined in the data dictionary. The mode (1 = Credit card) was checked and used as the replacement.

After all cleaning steps, the final dataset has 1,895,966 rows.

3. Exploratory Data Analysis

3.1 General EDA: Finding Patterns and Trends

Classify variables into categorical and numerical

Variable Classification:

Categorical (Nominal):

VendorID: encodes vendor identity (1 or 2), no numeric meaning
RatecodeID: rate type codes (1-6), no quantitative order
PULocationID: taxi zone identifier
DOLocationID: taxi zone identifier
payment_type: payment method code (1-6)

Datetime:

tpep_pickup_datetime: trip start time
tpep_dropoff_datetime: trip end time

Numerical – Discrete:

passenger_count: integer count of passengers

Numerical – Continuous:

trip_distance, fare_amount, extra, mta_tax, tip_amount,
tolls_amount, improvement_surcharge, total_amount,
congestion_surcharge, airport_fee, trip_duration

Ordinal (treated as numerical in analysis):

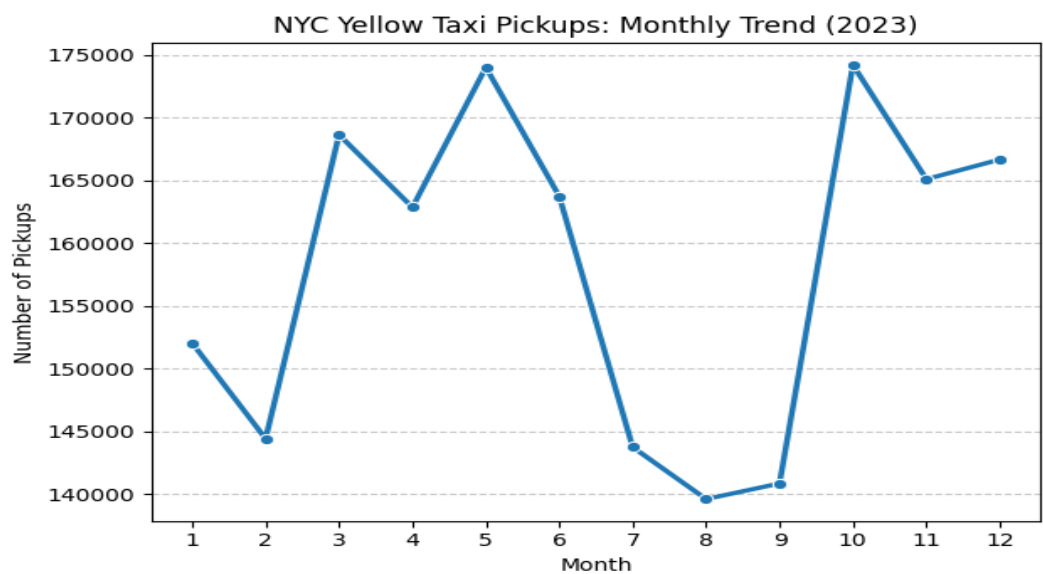
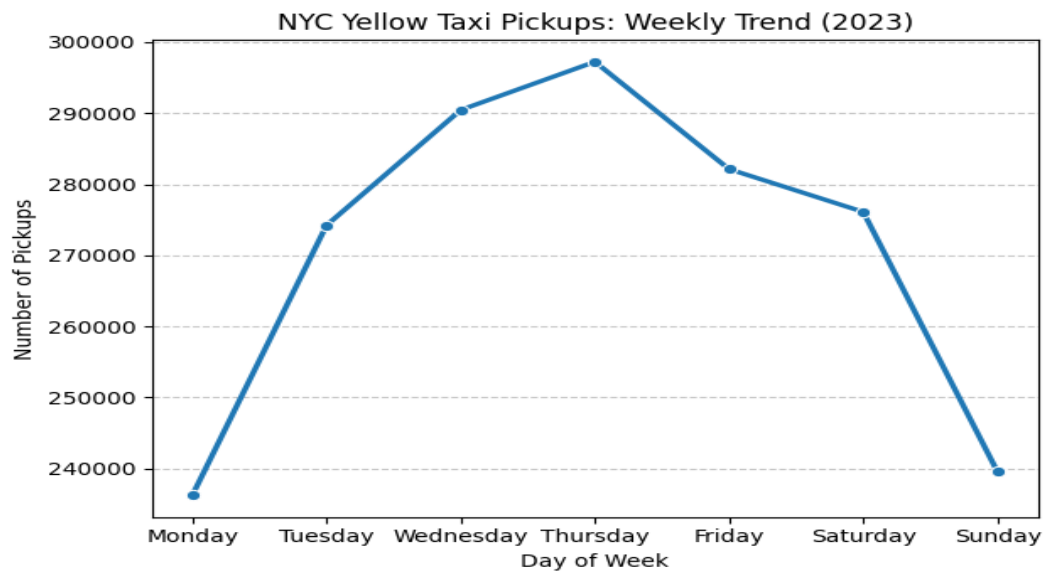
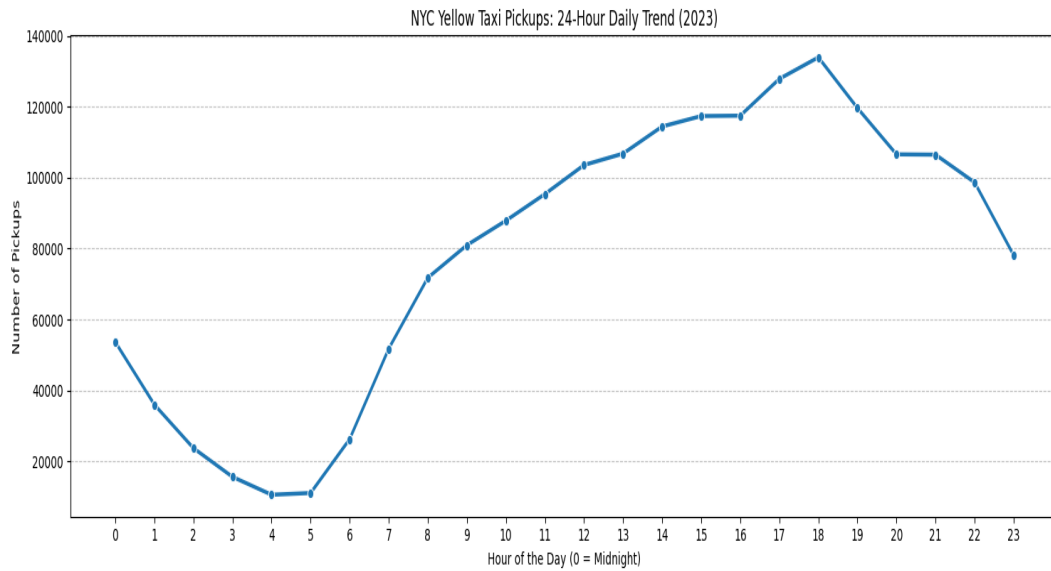
pickup_hour: hours 0-23, natural order

Analyse the distribution of taxi pickups by hours, days of the week, and months

Hourly trend: Demand follows a clear daily cycle. The quietest period is 3-5 AM and pickups build steadily from 6 AM onwards. Evening hours (6-8 PM) are consistently the busiest, likely driven by the post-work commute. There's also a noticeable late-night spike on weekends.

Daily trend: Friday has the highest trip count among weekdays. Sunday is the quietest. Saturday follows the weekday volume but with a different hourly shape, with demand peaking later and staying elevated past midnight.

Monthly trend: Trip volume is relatively stable across the year. January and February are the slowest months (possibly weather-related), and there are mild peaks in May and October.



Filter out the zero/negative values in fares, distance and tips

A filtered copy of the dataframe (`df_financial_analysis`) was created keeping only rows where `fare_amount > 0` AND `total_amount > 0`. These represent actual paid trips and are used for financial analysis.

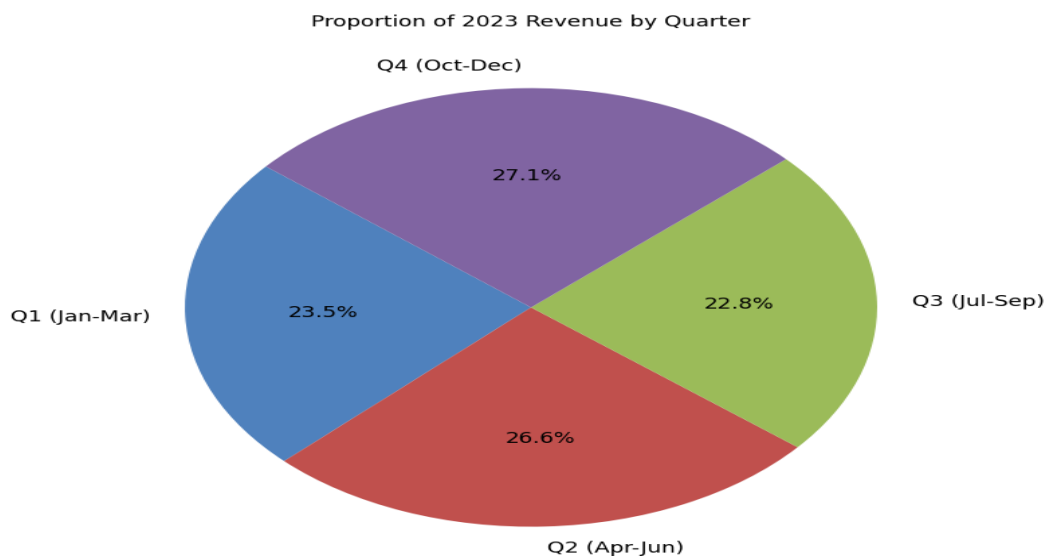
Zero `trip_distance` trips were kept, since a pickup and dropoff in the same zone can still represent a valid short ride (e.g., a few blocks within Midtown). Dropping all zero-distance trips would remove legitimate data.

Analyse the monthly revenue trends

Monthly revenue closely tracks trip volume, no big surprises there. Revenue is highest in May and October, and lowest in January/February. The average fare stays fairly stable across months (around \$19-21), so the revenue variation is almost entirely driven by trip count rather than fare changes.

Find the proportion of each quarter's revenue in the yearly revenue

The quarterly breakdown shows a fairly even split across the year. Q1 is slightly lower (~22-23%) which aligns with the slower winter months. Q2, Q3, and Q4 are roughly equal at around 25-26% each.



Analyse and visualise the relationship between distance and fare amount

As expected, trip distance and fare amount have a strong positive correlation ($r = 0.94$). The scatter plot shows a fairly linear relationship, with fare increasing roughly \$2.50-3.00 per mile. There's some scatter at longer distances, likely from airport flat-rate trips and negotiated fares.

Analyse the relationship between fare/tips and trips/passengers

Trip duration vs fare ($r = 0.85$): Also a strong positive relationship. Longer trips in traffic cost more even when the distance isn't huge. Rush-hour rides are a good example of this.

Passenger count vs fare ($r = 0.04$): Essentially no correlation, which makes sense. NYC taxis charge per trip, not per person. A group of 4 pays the same meter rate as a solo rider.

Tip amount vs trip distance ($r = 0.59$): Moderate positive correlation. Longer, higher-value trips generate better tips in absolute terms, particularly for credit card payments where tips are recorded automatically.

Analyse the distribution of different payment types

Credit card is the dominant payment method at around 65% of trips. Cash accounts for roughly 33%. Dispute, No Charge, and Voided trip categories are all marginal. One thing worth noting: cash tips aren't recorded in the data at all, so the tip analysis in later sections is essentially credit-card-only.

Load the taxi zones shapefile and display it

The `taxi_zones.shp` shapefile was loaded using GeoPandas. It contains 263 taxi zones covering all five NYC boroughs, with columns for `LocationID` (used to join with trip data), `zone name`, `borough`, and the geometry needed for map rendering.

Merge the zone data with trips data

Trip records were merged with zone data on `pulocationid = LocationID` (left join). This adds zone name and borough for each pickup location. The same merge was done for dropoff zones separately.

Find the number of trips for each zone/location ID

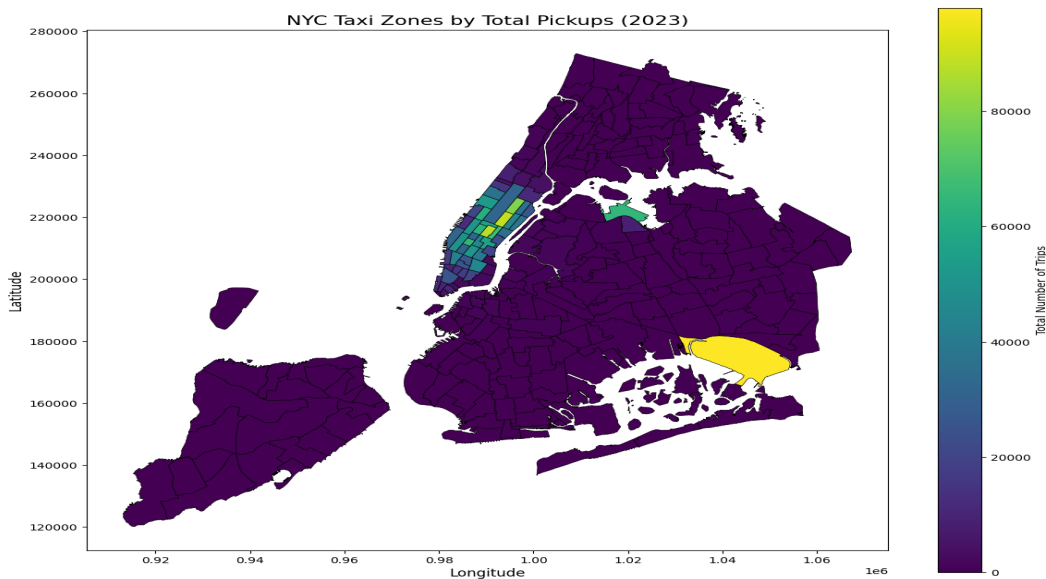
Trips were grouped by `pulocationid` and counted. Midtown Center leads by a large margin with close to 100k pickups in the sampled data, followed by JFK Airport and Upper East Side South.

Add the number of trips for each zone to the zones dataframe

Trip counts were merged back into the zones GeoDataFrame on `LocationID`. Zones with no pickups in the sampled data were filled with 0. This prepared the GeoDataFrame for the choropleth map.

Plot a map of the zones showing number of trips

The map makes the pattern obvious: Manhattan dominates completely. The zones around Midtown, Upper East Side, and the airport corridors are the darkest on the map. Most of the outer borough zones are barely visible, which tells us NYC yellow taxi activity is heavily concentrated in Manhattan.



Conclude with results

Summary of key findings from the general analysis:

Demand peaks at 6-8 PM on weekdays and late-night on weekends. Fridays lead in volume.

Monthly revenue tracks trip count closely, no pricing anomalies.

Q1 is the slowest quarter; Q2-Q4 are roughly balanced.

Distance is the strongest predictor of fare ($r = 0.94$). Trip duration is also strong ($r = 0.85$).

Passenger count has no effect on fare as expected.

Credit card dominates payment (~65%). Cash tips are not captured.

Manhattan (especially Midtown) accounts for the vast majority of pickups. Airport zones are next.

3.2 Detailed EDA: Insights and Strategies

Identify slow routes by comparing average speeds on different routes

Trip speed was calculated as $\text{trip_distance} / (\text{trip_duration_minutes} / 60)$ for trips with valid durations (1-180 minutes). The slowest route-hour combinations tend to appear in cross-borough routes during peak hours, where average speed drops to near 0 mph on some Zone-to-Zone pairs.

Overall city-wide average speed dips noticeably during morning (7-9 AM) and evening rush (5-8 PM), confirming the expected congestion windows. Pre-positioning cabs outside known slow corridors before peak hours would be more efficient than dispatching reactively.

Calculate the hourly number of trips and identify the busy hours

Looking at the hourly trip counts, 6 PM (18:00) is clearly the busiest hour in the sampled data. The chart shows a gradual build from morning and a sharp peak in the evening commute window, dropping off after 9 PM.

Scale up the number of trips from above to find the actual number of trips

Since the data is a 5% sample, actual trip counts are estimated by multiplying sampled counts by 20. The top 5 busiest hours (estimated actual trips):

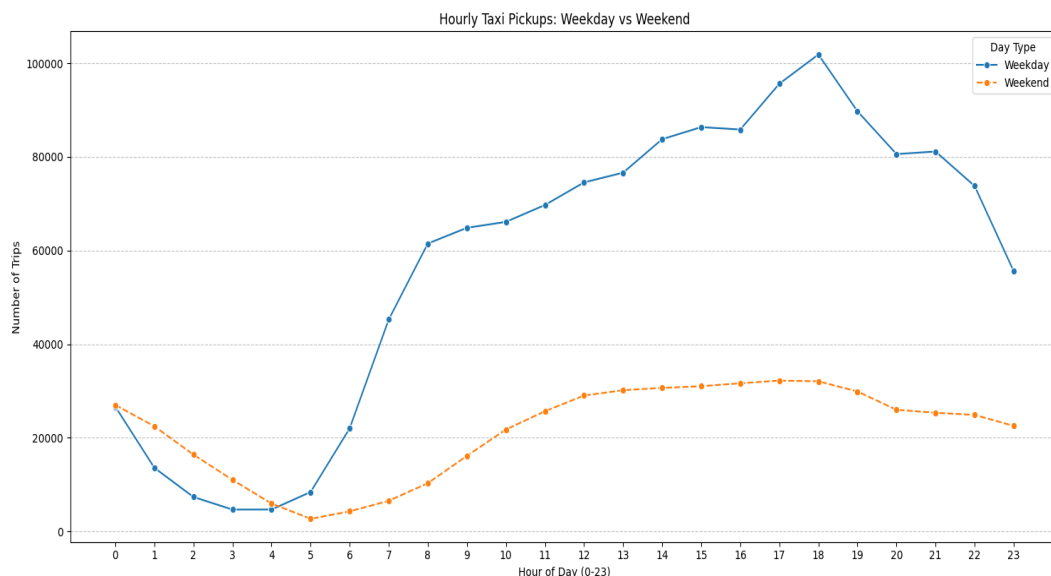
18:00: ~2,678,360 trips
17:00: ~2,556,840 trips
19:00: ~2,392,680 trips
16:00: ~2,349,560 trips
15:00: ~2,347,420 trips

All five are in the afternoon/evening window (3 PM - 7 PM).

Compare hourly traffic on weekdays and weekends

Weekdays show a clear twin-peak pattern: morning rush (7-9 AM) and evening rush (5-8 PM), driven by commuters. Weekends look quite different: demand builds gradually from around noon and peaks in the evening, staying elevated until past midnight. The late-night window on weekends is noticeably busier than on weekdays.

This suggests fleet deployment strategy should differ between weekdays and weekends rather than applying the same schedule to both.



Identify the top 10 zones with high hourly pickups and drops

Midtown Center at 6 PM has the highest hourly pickup count in the sampled data (7,680 trips). JFK Airport at 4 PM comes in third, showing the strong afternoon departure wave. Upper East Side South appears multiple times in the top 10 across different hours.

The hourly trend chart for these zones shows each has a distinctive peak time. Midtown peaks sharply in the evening while airport zones have a broader mid-afternoon peak.

Find the ratio of pickups and dropoffs in each zone

Zones with high pickup/dropoff ratios are 'origin' zones, where trips start more often than they end. JFK Airport has a ratio of ~4.2, meaning for every dropoff, there are 4+ pickups. LaGuardia is similar. These zones generate outbound trips and cabs arriving at these airports often have to wait for a return fare.

Zones with low ratios are 'destination' zones, where people arrive but don't leave by taxi. Understanding this imbalance helps with repositioning strategy.

Identify the top zones with high traffic during night hours

During night hours (11 PM - 5 AM), the pickup hotspots shift from business/commuter zones to entertainment areas. East Village tops night pickups, followed by JFK Airport (red-eye flights), West Village, and Lower East Side. These are all nightlife-heavy or transport-hub zones.

Dropoffs during night hours are spread more broadly, with riders going home to residential zones across all boroughs.

Find the revenue share for nighttime and daytime hours

Daytime hours (6 AM - 10 PM) account for 87.7% of sampled revenue (\$48.2M), while nighttime hours (11 PM - 5 AM) contribute 12.3% (\$6.8M). This is roughly proportional to the number of trips in each window. Average fares are slightly higher at night due to the overnight surcharge.

For the different passenger counts, find the average fare per mile per passenger

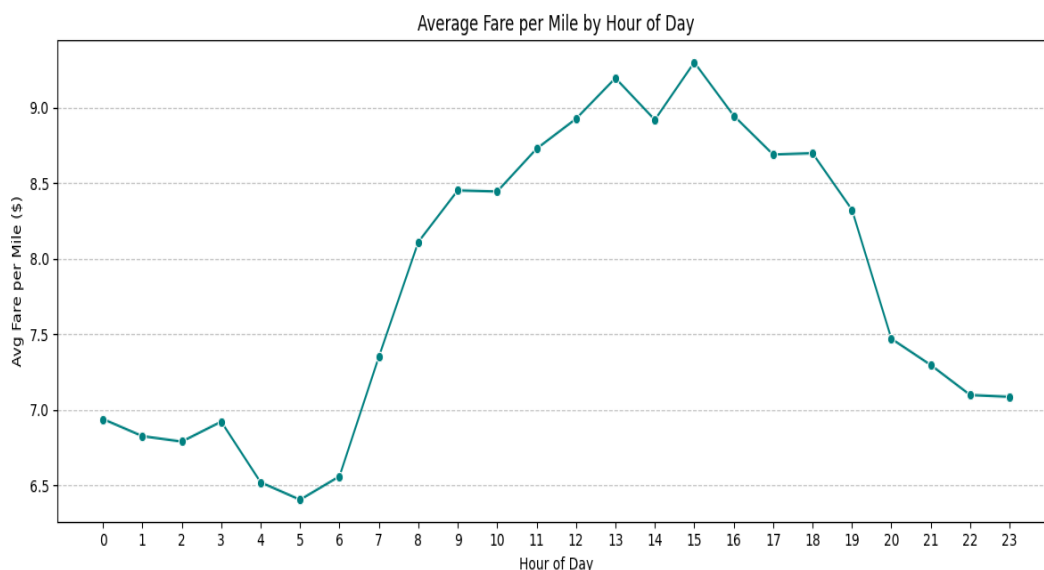
As passenger count increases, the fare per mile per passenger drops significantly. Solo riders pay around \$9-10 per mile in total fare. A group of 4 sharing that same fare comes out at roughly \$2.50 per person per mile. NYC taxis don't charge per-person, so larger groups get much better value per passenger.

This is worth noting for pricing strategy: a group ride discount could attract more group bookings while still maintaining overall per-trip revenue.

Find the average fare per mile by hours of the day and by days of the week

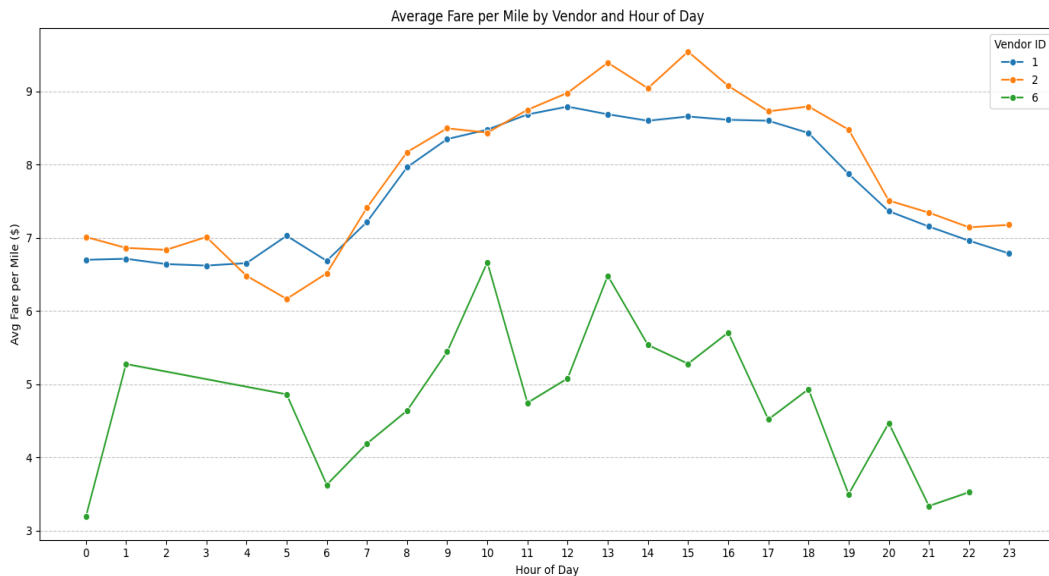
Fare per mile is highest during very early morning hours (1-4 AM) and dips during the day, reflecting both supply/demand dynamics and the composition of trips at those hours. Late-night trips tend to be longer and faster, covering more distance per dollar.

By day of week, Fridays and Saturdays show slightly higher fare per mile compared to mid-week days. The difference isn't dramatic, but it's consistent.



Analyse the average fare per mile for the different vendors

CMT (Vendor 1) and VeriFone (Vendor 2) are very close in fare per mile throughout the day. There's a small divergence in the late-night hours (1-4 AM) where VeriFone edges slightly higher. Overall, both vendors operate at near-identical pricing, which suggests the rate structure is standardised across both systems.

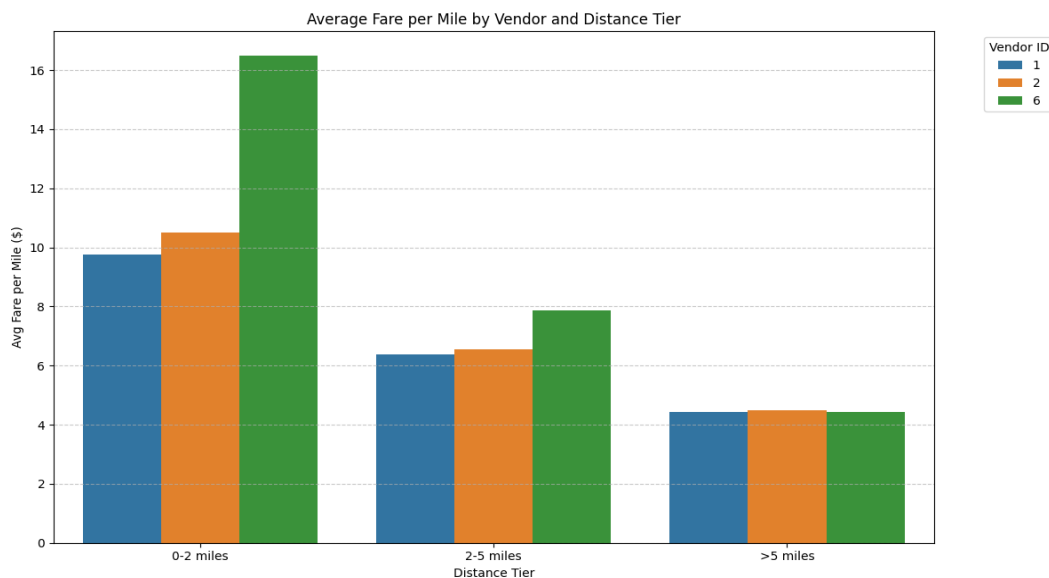


Compare the fare rates of different vendors in a distance-tiered fashion

Short trips (0-2 miles) command the highest fare per mile, around \$9.77 for CMT and \$10.50 for VeriFone. This makes sense: the flag-drop fee and minimum fare dominate short trips.

Medium trips (2-5 miles) drop to around \$6.40-6.54 per mile. Long trips (5+ miles) converge to around \$4.43-4.49 per mile for both vendors, so the rate structure effectively equalises at distance.

Both vendors show the same pattern, confirming that pricing is largely standardised by the NYC rate card rather than vendor-specific.



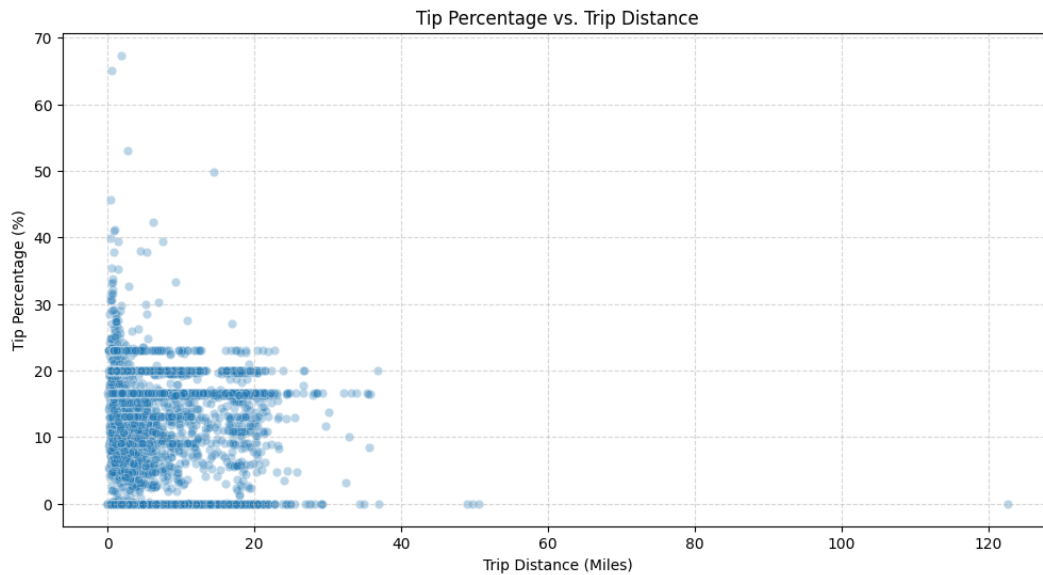
Analyse the tip percentages

Tip percentage analysis was done on credit card trips only since cash tips aren't recorded. The scatter plot shows a lot of zero-tip trips at short distances, where riders on quick hops don't tip much.

Comparing low-tip trips (<10%) vs high-tip trips (>25%): interestingly, high-tip trips actually have a lower average fare (\$14.63) and shorter average distance (2.44 miles) than low-tip trips (\$20.64 fare, 3.73 miles). This

suggests tip percentage is driven more by payment behaviour and personal habit than by trip value.

Late-night and early-morning hours show lower average tip percentages. Short trips in these windows with cash payment are likely the main drivers of low tips.



Analyse the trends in passenger count

Solo trips dominate across all hours and days, roughly 76% of all trips have 1 passenger. Average passenger count is slightly higher on weekend evenings and early mornings, where group travel (dining, nightlife) is more common.

The heatmap shows the variation is fairly small. The difference between the highest and lowest average passenger count across hour/day combinations is less than 0.5 passengers.

Analyse the variation of passenger counts across zones

Airport zones (Newark, JFK, LaGuardia) and some outer borough zones have higher average passenger counts, as people travelling to/from airports more often travel in groups. Midtown zones tend toward solo riders.

Analyse the pickup/dropoff zones or times when extra charges are applied more frequently

Surcharge application across the dataset:

MTA tax: 99.1% of trips (essentially universal)

Improvement surcharge: 100.0% of trips (universal, applied at flag drop)

Congestion surcharge: 89.2% (very common, applied in most Manhattan trips)

Extra (rush/overnight): 59.9% (over half of trips)

Airport fee: 7.8% (JFK and LaGuardia only)

The congestion surcharge is applied consistently throughout daytime hours and drops off significantly between 12 AM and 6 AM, which aligns with when the NYC congestion pricing zone is active.

4. Conclusions

4.1 Final Insights and Recommendations

Recommendations to optimize routing and dispatching based on demand patterns and operational inefficiencies

From the hourly and weekly patterns, weekday evenings around 6-8 PM are consistently the busiest period. The speed analysis confirmed this. Average speed drops noticeably during those hours, so pre-positioning cabs 20-30 minutes before the rush would work better than reacting after demand spikes.

A few specific things stand out:

Weekend dispatch is different, demand builds slowly and stays high until past midnight. Running the same schedule as weekdays wastes morning coverage and leaves gaps late at night. A separate weekend shift pattern makes more sense.

Airport zones have a rebalancing problem. The pickup/dropoff ratio showed JFK (4.2x) and LaGuardia (3.7x) have way more dropoffs than pickups. Drivers finishing airport trips should head back to Midtown rather than queue at the airport, since the pickup wait there isn't worth it most of the time.

Slow routes during peak hours are concentrated in Midtown corridors. Routing cabs through cross-streets or staging them slightly outside congestion zones before rush hour would reduce idle time.

Suggestions on strategically positioning cabs across different zones

The choropleth map made it clear that almost all pickup activity is in Manhattan, with Midtown, Upper East Side, and Penn Station area accounting for the bulk of it. That pattern is consistent across all months, so these zones should always have good coverage.

What changes by time of day:

On weekday mornings, cabs should be concentrated near residential blocks in Upper East Side and transit hubs since people are heading to offices in Midtown and the Financial District.

Night hours (11 PM-5 AM) shift the hotspots to entertainment areas. Lower Manhattan and Midtown West see more activity, especially on Fridays and Saturdays. Keeping some coverage there overnight on weekends is worthwhile.

January and February are the slowest months. Running slightly fewer active cabs then would cut idle time without hurting service. May and October are the peaks, so scaling up for those months makes sense.

The outer boroughs are mostly underserved in the current data. There's room to grow there but it would need targeted driver incentives.

Propose data-driven adjustments to the pricing strategy

A few things stood out from the pricing analysis:

Short trips under 2 miles have the highest fare per mile (around \$9-10) but passengers on those trips tend to tip the least. Longer trips have lower per-mile rates but bring in better tips overall. Adding a minimum short-trip charge would help recover the overhead on quick rides (boarding time, traffic stops) without being too aggressive.

The vendor comparison (CMT vs VeriFone) shows they're very close in pricing across all distance tiers. Both converge to around \$4.43/mile for trips over 5 miles. There's no significant competitive differentiation in pricing.

The biggest insight from the tip analysis was payment type. Credit card trips recorded much higher tips than cash. Since cash tips aren't captured at all, the actual gap may be smaller, but encouraging cashless payment (even with a small fare discount) would improve recorded tip revenue and likely driver retention.

For peak hours (weekday evenings and Friday/Saturday nights), a modest surge of around 10-15% would reflect the actual demand without driving riders away. The fare per mile data already shows higher average fares during those windows, so the market is already bearing it.